

13.6 A 16Gb 37Gb/s GDDR7 DRAM with PAM3-Optimized TRX Equalization and ZQ Calibration

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GDDR memory has consistently maintained a leading position in delivering high performance: necessary for applications such as artificial intelligence, deep learning, and data centers. However, achieving an elevated I/O bandwidth, beyond the latest 27Gb/s GDDR6 [1], presents significant challenges in ensuring a sufficient I/O link budget. One solution is using multi-level pulse amplitude modulation (PAM) signaling. In the GDDR7 I/O specification, single-ended PAM3 signaling is used to achieve higher data rates as it can transfer data more than 58% faster, compared to non-return-to-zero (NRZ) signaling. Furthermore, PAM3 offers a 50% larger voltage sensing margin than PAM4, which was adopted for GDDR6X [2]. Therefore, PAM3 exhibits a higher potential for bandwidth extension, given that single-ended signaling is vulnerable to random noise and crosstalk.

This paper presents a 16-Gb GDDR7 DRAM with PAM3 signaling in a 1.2V DRAM process; it achieves 37Gb/s and a 20% improvement in energy efficiency, compared to prior GDDR6 DRAM. The TX is implemented with an adaptive gain-controlled feedforward equalizer (FFE) to optimize performance over process, voltage, and temperature (PVT). The RX uses a continuous-time linear equalizer (CTLE) and a 1-tap decision feedback equalizer (DFE), where the two separated CTLEs are merged with current summer DFE to simplify the feedback path. Additionally, the proposed ZQ calibration scheme uses alternately switched reference voltages to enhance the level mismatch ratio (RLM) of the PAM3 signaling. Moreover, low-power and low-jitter clocking techniques are employed for WCK distribution. A low-dropout (LDO) regulator in the command clock (CK) path reduces the power-supply-induced jitter (PSIJ). CMOS clocking for the DQ clock path improves power consumption during read and write in low frequency operation.



The overall architecture of the proposed GDDR7, consisting of 2 dies for 4-channel configurations, is shown in [Fig. 13.6.1](#). Each die operates as a 2-channel for 4-channel mode or a 1-channel for 2-channel mode. Each channel has 11 DQs in PAM3 format, which undergo an 11b to 7-symbol (11b7s) encoding or decoding process for read and write operation in DRAM cells. With a similar WCK frequency, the total bandwidth of 11 DQs in GDDR7 is equivalent to the $2\times$ bandwidth of 8 DQs in GDDR6. RCK_t/RCK_c are added as read data strobes on the host side to improve the timing margin by compensating for correlated jitter. The number of CA pins in GDDR7 is reduced, compared to GDDR6, owing to the quad data-rate command operation. The clock for the command is internally generated using a quarter rate of the WCK frequency.

[Figure 13.6.2](#) illustrates the implementation of the proposed PAM3 TX, which includes two pairs of 4:1 multiplexers, an on-chip equalizer, a main driver, and an FFE driver. Because the input/output capacitance (CIO) is critical for PAM3 signaling, a non-stacked driver with a passive resistor is used instead of the double-stacked driver in [\[1\]](#); as a result, CIO is decreased by 28%. Also, a fractionally-spaced FFE, suitable for multi-level PAM signaling [\[3\]](#), is implemented. The FFE coefficients are controlled by an inverter-based digitally-controlled delay line and gain controllable slicers. To optimize the FFE performance, we used an encoder that adaptively adjust the delay and strength based on regularly updated ZQ calibration results. In the fast process corner, optimal FFE coefficients are achieved by increasing the delay and decreasing the enabled slicers; whereas, in the slow corner, the delay is decreased and the number of enabled slicers is increased. In a similar manner, voltage and temperature drift are also managed by the encoder. Consequently, the proposed FFE can maintain robust performance over the PVT variation.

The proposed PAM3 RX architecture consists of a CTLE and DFE; two reference voltages are used, which allows the two separated CTLEs to be individually optimized for the upper and lower eyes. The proposed PAM3 DFE is implemented using a current summer type; 4-phase feedback paths are combined using the voltage-to-current converter (V2I), as shown in [Fig. 13.6.3](#). As a result, the DFE-feedback timing margin can be improved by lightly loading the sense-amplifier (SA) outputs, because the number of global DFE feedback lines can be reduced by $\frac{1}{4}$ compared to a conventional direct feedback DFE [\[1\]](#). Furthermore, instead of using a CML buffer in the current-summing DFE, the CTLE serves as the DFE summer; hence, overall power consumption is reduced.

The amount of inter-symbol interference (ISI) is different in each signal level for single-ended PAM3. Hence, the proposed DFE tap coefficient is independently applied to the upper and lower eyes via GAIN_H and GAIN_L, depending on the previous symbol. Simulation results show that, regardless of the DFE gain strength, the proposed DFE symmetrically improves the upper and lower voltage margin for the both eyes compared to a conventional PAM3 DFE [\[4\]](#). Additionally, each DQ has total 8 SAs; the mismatches of those sampling transistors are calibrated by dedicated offset-cancelling controller in each SA during DRAM initialization.

The operation of the proposed PAM3 ZQ calibration is depicted in [Fig. 13.6.4](#). To transmit the three PAM3 symbol levels, denoted as +1, 0, and -1, each pull-up and pull-down driver is divided into two separate branches. Due to the inherent MOSFET non-linearity, the impedances of both pull-up and pull-down drivers vary with the symbol level, degrading the RLM. Therefore, the proposed ZQ calibration uses two reference voltages to mitigate this non-linearity and achieves a RLM improvement compared to the conventional ZQ calibration in [\[1\]](#). The reference voltage in the calibration loop is switched at regular intervals according to the all-bank refresh (ABR) command. The reference voltage of $0.75 \cdot V_{DDQ}$ is used to set the level of symbol 0, by adjusting the pullup code PU_B[n:0] and the pull-down code PD_A[n:0]. The other pull-up code PU_A[n:0] is encoded from PU_B[n:0] to target the impedance for symbol +1. Next, $0.5 \cdot V_{DDQ}$ is used to calibrate the -1 symbol level. The impedance mismatch of the pull-down branch by PD_A[n:0] for $0.5 \cdot V_{DDQ}$ level is compensated by adjusting the other pull-down code PD_B[n:0]. Additionally, this process also derives the on-die termination (ODT) code PU_ODT[n:0] from the combination of calibrated pull-up codes. As a result, the proposed ZQ calibration improves the RLM by 3.4% and alleviates the mismatch between upper and lower eye sizes.

The overall WCK distribution of the proposed GDDR7 is shown in [Fig. 13.6.5](#). In the prior GDDR6, the I/O and command used separated clocks, hence the CK path could be matched with the CA path. However, we implemented a CK path that is internally generated by WCK. In other words, an unmatched CA path is employed, where the CK path is much longer than the CA path. Consequently, the uncorrelated jitter, including PSIJ between the CK and CA paths, degrades CA timing margin. To mitigate PSIJ, the regulated voltage is generated by an LDO is applied to the CMOS repeaters in the CK path. A regulated voltage of 0.85V

from V_{DDQ} is determined to guarantee high-speed operation of CMOS buffers, and to satisfy the operating condition of the LDO for lower V_{DDQ} as well. As a result, the proposed GDDR7 achieves the PSIJ of 1.6ps/10mV in the CK path and it can operate with a 1.1V supply voltage. In addition, to reduce the power consumption in low-frequency mode, a CMOS clock distribution is used, instead of CML clock distribution, for the DQ clock path. The CMOS and CML clock trees are merged by the CML-to-CMOS (C2C) circuitry. Consequently, the proposed WCK distribution reduces power consumption by more than 60mA per die under 16Gb/s operation, without degrading jitter performance in high frequency operation.

The measurement results for the all-DQ merged TX eye diagram and RX Shmoo plot at a 32Gb/s data rate are shown in [Fig. 13.6.6](#). A pseudo-random pattern is used after perpin training. The TX FFE's upper and lower eye openings are 130mV/29ps (0.46UI) and 110mV/26ps (0.41UI). The RX equalization achieves a valid window with 132mV/33ps (0.53UI) and 144mV/33ps (0.53UI). The RLM is measured to be 0.982 during 4Gb/s operation. With a gap-less read and write operation, the measured t_{WCK} Shmoo plot indicates that the maximum 37.2Gb/s operation speed is achieved with 1.2V, and 34Gb/s at 1.1V. [Figure 13.6.7](#) shows a photograph of the proposed GDDR7 that occupies 41.6 mm² and is fabricated with a 1z DRAM technology.



Figure 13.6.1:

Proposed GDDR7 architecture compared to GDDR6 (top) and block diagram of GDDR7 interface (bottom).



Figure 13.6.2:

PAM3 transmitter with adaptive gain-controlled FFE (left) and simulation results (right).



Figure 13.6.3:

Conventional direct feedback DFE (top-left) and proposed DFE (top-right). The proposed DFE tap coefficient and simulation results for the RX Shmoo plot (bottom).



Figure 13.6.4:

Driver operation for each symbol (top-left). RLM improvement by the proposed ZQ calibration (top-right). ZQ calibration operation using each reference voltage (bottom).



Figure 13.6.5:

WCK distribution for DQs and CAs (top). Implementation of DQ C2C (bottom).



Figure 13.6.6:

All-DQ merged measurement results for TX eye diagram and RX Shmoo plot at 32Gb/s (top). Measured RLM (bottom-left) and t_{WCK} Shmoo plot (bottom-right).

Figure 13.6.7:
Chip micro photograph and summary table.

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